Spinning and Boosted Trumpet Initial Data: Formulation and Validation Status

TwoPuncturesC and AthenaK integration

(Dated: 18 October 2026 — interim report)

Abstract

Selectable trumpet and quasi-isotropic (QI) seeds, the coupled solver, portable checkpoints and the AthenaK consumer are implemented. **No complete binary physical-validation claim is supported yet.** A focused QI control with the same generic moderate binary seed parameters shows that widening the interior transition reduces coarse-grid bulk momentum RMS by a factor of 36.4. One unchanged QI formulation at three resolutions then meets every absolute exterior constraint bound, but its bulk residuals increase with refinement; the original strict trend criterion remains failed. Both extreme trumpet binaries also remain unqualified. The regularity investigation distinguishes puncture, transition, axis and compactified-infinity behavior; a seed-family change alone is not a demonstrated general remedy. The finest QI checkpoint passes sampler migration and horizon angular/quadrature checks, with its failed constraint-trend gate retained separately. This report records an incomplete validation milestone.

I. CONSTRUCTION AND ITS LIMITS

The reference is the $R_0 = M$ stationary Kerr trumpet of Dennison, Baumgarte and Montero, Eq. 12 of arXiv:1409.1887v2. The reused AthenaK reference is pinned to `project/kerr-trumpet-spin09`, commit `81c7d8030c49489577c54b977b8b7f7b42d9a96e`. This is distinct from both Ian Ruchlin's quasi-isotropic wormhole HiSpID free data and a stationary standard $1 + \log$ trumpet. The existing quasi-isotropic family remains the default; residual-Z4c evolution background subtraction is not added to the elliptic equations.

We use signature $(- + ++)$ and $K_{ij} = -\mathcal{L}_n \gamma_{ij}/2$. For rest mass m, $a = |\boldsymbol{S}|/m$, and $R = r + m$, one scalar choice is $\psi_0 = (\Sigma/r^2)^{1/4}$ with $\Sigma = R^2 + a^2 \cos^2\theta$; the other is $(\det\gamma)^{1/12}$. Both have direction-dependent $r^{-1/2}$ puncture behavior. Full spacetime Lorentz pullbacks supply boosted spatial tensors and their time derivatives. A rest-chart connection contraction gives the extrinsic curvature stably. Spacelikeness and positive spatial metrics are checked without clipping. Nonspinning boosted slicing has an analytic global spacelikeness argument; general spinning cases retain sampled-domain checks.

The four elliptic unknowns are a scalar correction u and vector b^i. Outside the explicitly

modified interior, the CTT equations are

$$\widetilde{\Delta}\psi - \tfrac{1}{8}\widetilde{R}\psi - \tfrac{1}{12}K^2\psi^5 + \tfrac{1}{8}\widetilde{A}_{ij}\widetilde{A}^{ij}\psi^{-7} = 0,$$
$$\widetilde{D}_j\widetilde{A}^{ij} - \tfrac{2}{3}\psi^6\widetilde{D}^i K = 0, \qquad \widetilde{A}^{ij} = \widetilde{A}_0^{ij} + (\widetilde{L}b)^{ij}.$$

The covariant CTT tensor is $\widetilde{A}_{ij} = \psi^2(K_{ij} - \gamma_{ij}K/3)$; AthenaK's Z4c conformal tensor instead uses ψ^{-4} on the same physical trace-free tensor. The import explicitly converts these conventions.

Unmodified trumpet corrections cannot simply be assumed analytic at the puncture: the isolated Schwarzschild scalar indicial exponent includes $p_0 = \sqrt{3} - 1/2$. The implemented binary examples use the declared interior plateau, where the correction equations become $\Delta u = 0$ and $\Delta b + \nabla \operatorname{div} b/3 = 0$. Bounded corrections then have removable singularities. This modifies the interior operator; vacuum accuracy is tested independently outside it, and measured horizons must enclose the modified regions. It is not evidence of exponential convergence across the transition.

II. ACCEPTANCE AND PRODUCTION PATH

The production stages are host extended-precision geometry/cache construction, CUDA GMRES for the solve, CPU sampling, and OpenMP AthenaK horizon integration. The opt-in no-swirl axisymmetric Krylov projection is restricted to nonspinning axial configurations; full unprojected nonlinear residuals still govern acceptance. Default QI and Bowen–York paths are preserved. No backend benchmark matrix is part of this investigation.

Obligation	Decisive criterion	Status
Seeds and boosts	Independent fields/derivatives, vacuum checks, charges, positive slicing; generic orientation included	Pass within tested domain
Operators	Manufactured curved operators and centered JVP checks	Pass
Binary constraints	Near and bulk RMS $< 10^{-6}$, maximum $< 10^{-4}$; three-grid decrease, positive ψ, stable charges	Not passed
Import	Checkpoint-bound fields and gradients within 10^{-12}	Passed cases
Horizons	Expansion RMS $< 10^{-7}$; angular/quadrature stability, mass/spin changes $< 10^{-4}$; enclosure	See below
Compatibility	Unchanged QI default fixture on identical platform/build	Pass

The physical observers are independent of the weighted collocation residual. Modified-region errors are recorded separately. A successful solve or import does not promote diagnostic data to physical acceptance. Checkpoints preserve family, basis/map and producer identity; v1 retains its QI meaning and v2 adds explicit family metadata.

III. LATEST REGULARITY INVESTIGATION: 18 OCTOBER

The unmodified Schwarzschild scalar sector admits $u \sim r^{\sqrt{3}-1/2}$, so factoring out the seed singularity does not imply an analytic correction. The flat interior plateau removes that local scalar potential, but not possible regularity limitations at its nonanalytic compact transition or at infinity. This isolated scalar exponent is not the full coupled spinning/boosted indicial spectrum.

Offline analysis of retained $160 \times 320 \times 16$ factored $m = 0$ unknowns finds slow meridional coefficient tails in both seed families. Degree-eight polynomial windows localize the wider-interior trumpet's radial vector tail mainly toward compactified infinity: suppressing that endpoint leaves 0.00406 of the degree 80–143 band norm, whereas suppressing the inner radial boundary leaves 1.003. The narrow QI control has the opposite localization. The inner boundary includes both foci and the axis segment between them. These are representation diagnostics of distinct interiors, not a controlled slicing-only comparison or an identification of logarithmic powers. No filtered field is used as a solution.

The QI control retains masses 0.6/0.4, separation 6, and the original generic spin and velocity vectors ($|\chi| = 0.3905/0.3742$). It uses the exact seed scalar choice, no far attenuation, the same mapped modal basis, and the original modal GMRES on CUDA. The narrow interior window spans fractions 0.15–0.5 of the isolated QI throat radius. Its 160-grid solve converges internally in 8 Newton and 46 Krylov iterations, but bulk momentum RMS 3.809×10^{-6} fails the 10^{-6} bound. A separate candidate widens the outer fraction to 0.85, retaining the 0.15 plateau. At unchanged exterior points this reduces bulk momentum RMS to 1.045×10^{-7}. Its fixed sequence is:

Grid	Near H	Near M	Bulk H	Bulk M	Krylov
$160 \times 320 \times 16$	3.13×10^{-7}	9.14×10^{-8}	6.30×10^{-9}	1.05×10^{-7}	35
$192 \times 384 \times 16$	1.44×10^{-7}	4.29×10^{-8}	8.62×10^{-9}	1.09×10^{-7}	34
$224 \times 448 \times 16$	4.88×10^{-8}	2.03×10^{-8}	5.32×10^{-8}	1.70×10^{-7}	251

All three use the same native image and physical free data, and all pass the absolute near/bulk RMS and maximum bounds. The 48 near-hole and 18 bulk points are fixed, including all finite-difference stencil points in the unmodified region. The nonlinear weighted maximum is below 10^{-12} on every grid; ψ and spatial metrics are positive at the declared observers. Extrapolated charge changes, scaled by $1+|Q_{\text{finer}}|$, are 3.83×10^{-10} and 2.35×10^{-9}, below 10^{-3}. **Bulk errors do not decrease, so this sequence fails the original strict convergence criterion.**

The physical norms are equal-point RMS values of $H = {}^{(3)}R + K^2 - K_{ij}K^{ij}$ and $|M|_\gamma = (\gamma_{ij}M^iM^j)^{1/2}$, with $M^i = D_jK^{ij} - D^iK$, in units M_0^{-2}, $M_0 = m_+ + m_- = 1$. They are not the internal sine-weighted conformal residual with endpoint rows. A supplemental combined exterior RMS $[\langle H^2 + |M|_\gamma^2\rangle]^{1/2}$ falls from 2.83×10^{-7} to 1.41×10^{-7} to 1.03×10^{-7}. This post-study summary does not replace the failed regional trend gate.

Necessary native solve times are 41.8, 52.7 and 140.3 seconds, after 60.6, 87.3 and 116.6 seconds of setup, with 8, 5 and 6 Newton steps. The first grid starts from zero; later grids reuse the preceding field as an initial guess. These are workflow records, not a matched-accuracy speedup measurement. No multigrid hierarchy is adopted: the retained stalled trumpet residual is predominantly grid-scale, and a coarse-error bottleneck has not been demonstrated. The bounded overintegrated, two-sweep modal and left-GMRES attempts also fail fine-grid linear targets; their optional implementations remain off by default.

The finest QI checkpoint passes separate-process producer/CPU-consumer sampling at 2454 points, with identical fields and metric gradients. The first AthenaK attempt stops before the finder because its Python proof validator recognizes only older basis identifiers. The compatibility fix explicitly admits C4 and axis-tau identifiers while retaining exact checkpoint, basis, map, image and field bindings; four regression tests pass, including rejection of unknown identifiers and invalid matching maps. The passed migration is reused, without a repeated sampling run.

AthenaK then passes harmonic orders $8, 12, 16$ and an independent quadrature increase from $n_\theta = 32$ to 48, with zero evolution. ADM/Z4c import error is 4.08×10^{-16}. The final measured properties are:

Hole	M_H	M_{irr}	χ_{coord}	Expansion RMS
$+$	0.615673713	0.604595771	0.370895278	2.04×10^{-8}
$-$	0.416086392	0.409618329	0.345813441	5.63×10^{-9}

All angular orders meet the 10^{-7} expansion bound. Across all consecutive angular and quadrature refinements, maximum mass-relative and dimensionless spin-vector changes are 2.00×10^{-12} and 2.78×10^{-11}. After the observed refinement buffer, the continuous modified-ball enclosure margins are 0.03205 and 0.01867. These are coordinate rotation-integral spins, not approximate-Killing-vector estimates, and angular stability does not measure the input solution's continuum error. Finder times are 47.5, 36.5, 38.4 and 43.8 seconds with 16 OpenMP geometry threads.

These import and horizon checks are diagnostic because the input sequence failed its strict bulk trend criterion. They neither qualify the trumpet family nor upgrade the QI checkpoint's diagnostic acceptance. Exact configurations, raw physical and horizon arrays, frozen-image hashes, figures and commands are retained in `validation/trumpet/regularity-family-control/`.

IV. RETAINED TRUMPET BINARY EVIDENCE

The retained pre-tau entries below are independent physical RMS errors, in the declared normalization. The moderate case has masses 0.6/0.4, separation 6, and generic spins and boosts; exact parameters are in `examples/trumpet_configs.py`. The spin target has equal

masses 0.5, aligned seed $\chi = 0.99$, and separation 12. The head-on target has opposite seed $\Gamma = 10$, masses 0.5, and separation 25. The latter has not yet been calibrated with qualified binary irreducible masses.

Case / grid	Near H	Near M	Bulk H	Bulk M
Moderate $192\times384\times16$	1.13×10^{-6}	3.77×10^{-7}	5.76×10^{-8}	4.42×10^{-7}
Moderate $224\times448\times16$	2.08×10^{-7}	1.65×10^{-7}	8.71×10^{-8}	4.44×10^{-7}
Moderate $256\times512\times16$	1.70×10^{-7}	1.39×10^{-7}	2.62×10^{-7}	2.86×10^{-7}
Spin $160\times320\times32$	3.15×10^{-6}	1.44×10^{-5}	8.30×10^{-5}	3.00×10^{-4}
Boost $256\times320\times8$	8.23×10^{-4}	2.42×10^{-2}	4.46×10^{-6}	1.27×10^{-4}
Boost $512\times320\times8$	1.02×10^{-4}	6.12×10^{-3}	1.07×10^{-6}	3.75×10^{-5}

The retained C2 moderate sequence fails because bulk errors do not decrease. A diagnostic removal of the scalar $m = 6$ pair reduces finest-grid bulk Hamiltonian RMS from 2.62×10^{-7} to 5.20×10^{-9}; the mode also has a forbidden ρ^4 axis term whose coefficient grows with resolution. This diagnoses sensitivity, not permission to filter a physical solution. Tighter stopping and smaller observer steps do not remove the defect.

A. Regularity correction and selected solver

The current opt-in construction retains C2 modal unknowns and imposes $P(0,\eta) = P(t,\pm1) = 0$ for $m \geq 5$ through independent endpoint rows. It avoids dividing sources by additional vanishing axis factors. A scalar manufactured case converges to 2.40×10^{-14}; under an absolute 10^{-16} source perturbation, its finest-grid field noise is 2.17×10^{-11} versus 3.43×10^{-6} with factored C4 unknowns. The forcing has an independent Cartesian-Hessian check. Native CPU/CUDA endpoint-row reconstruction, directional JVP and checkpoint controls pass; the largest CUDA row and JVP errors are 3.33×10^{-16} and 1.00×10^{-11}, respectively.

The preconditioner must enforce the same endpoint rows. Exact radial rows fit its existing dense blocks. Exact polar rows use a rank-$2n_A$ Woodbury update, with batched response construction and shared CPU/Kokkos application. An independent dense-matrix test including pivots and nonzero boundary data gives solution error below 6.7×10^{-15}. Small coupled solutions agree between CPU and CUDA to 2.82×10^{-13}; residual/JVP witnesses

are unchanged by the preconditioner modification. Allocation guards include the extra matrices.

The full $192 \times 384 \times 16$ moderate solve in job 59420346 now converges in 5 Newton/53 Krylov steps, taking 65.4 seconds after 65.5 seconds of setup. All Newton steps are undamped. The independent near-hole H/M RMS errors are $1.128 \times 10^{-6}/3.789 \times 10^{-7}$; bulk errors are $5.739 \times 10^{-8}/4.548 \times 10^{-7}$. Only the near-hole Hamiltonian RMS exceeds its bound. The workflow takes 246.6 seconds and peaks at 6.025 GiB host RSS. This grid remains diagnostic. The $224 \times 448 \times 16$ continuation initially fails a late linear solve, ending at relative residual 0.0245 after 2400 iterations against the requested 0.001. Resuming the saved iterate with standard forcing 0.1 takes one additional Newton step and 119 Krylov iterations (36.8 seconds), reaching nonlinear maximum 3.35×10^{-13} against the unchanged 10^{-12} tolerance. Independent near H/M RMS errors are $2.071 \times 10^{-7}/1.651 \times 10^{-7}$; bulk errors are $5.023 \times 10^{-8}/4.336 \times 10^{-7}$. All per-grid physical bounds pass. The subsequent 240×480 run with restart 24 fails a late linear solve (relative residual 0.873 after 2400 iterations); bulk momentum RMS is 1.72×10^{-6} and the nonlinear maximum is 1.62×10^{-9}. Resuming that saved state with restart 64 also fails after 2400 iterations: the true relative residual improves to 0.121, but exceeds the required 0.1. No Newton update is accepted. This retry takes 430.7 solve seconds and 692.8 workflow seconds; peak host RSS is 11.54 GiB. The allocation is released and the 256×512 level is not launched. Further work must diagnose the linear operator/preconditioner rather than relaxing the final criteria. The passing 224-grid result is retained. All levels use $n_\phi = 16$, the same native image, and unchanged nonlinear and physical acceptance criteria. Each completed grid must pass nonlinear and physical bounds before continuing, and the final assessment checks decreasing errors and charges. A matching CPU sampler and AthenaK consumer built successfully in job 59420409; new-checkpoint import and horizon measurements remain to be run. The option and distinct checkpoint representation remain experimental.

A bounded 64-step replay diagnoses genuine slow linear convergence: the true relative residual is 0.922, and recurrence versus explicit residual differ by only 1.97×10^{-13} relatively. A manufactured matrix witness at the failing resolution is recovered by all host modal factors to 1.12×10^{-9} or better. This checks host factor accuracy for that witness, not the approximation quality or the GPU inverse application. The subsequent actual-RHS comparison finds a 7.48% global GPU/host preconditioner discrepancy, dominated by mode-

4 momentum components. Before this correction, residual energy is 99.72% momentum and 70.92% modes 0–1; tau rows contribute negligibly. Feeding identical projected input to both inverse routines reduces the discrepancy to 2.98×10^{-16}, isolating Fourier projection before near-axis row scaling. An opt-in compensated projection preserves subtraction and product rounding errors during summation. Exact dyadic cancellation controls pass on the host; the small nonlinear control retains five Newton/53 Krylov iterations and changes sampled fields by only 3.12×10^{-13} (scaled), with bitwise unchanged residual/JVP witnesses. GPU cancellation controls also pass, and the corrected actual-state GPU/host inverse difference is 2.82×10^{-16}. A full Newton step now converges in 319 Krylov iterations, but the next inner solve stops at relative residual 0.306 after 2400 iterations. The nonlinear maximum improves to 2.17×10^{-11} but still fails 10^{-12}; bulk momentum RMS 2.53×10^{-6} also fails. A continuation with inexact-Newton forcing $\eta = 0.5$ accepted one step but the next linear solve stalled at relative residual 0.547 after 2400 iterations. Bulk momentum RMS remained 2.53×10^{-6}; this continuation also fails acceptance. For $Js = -r + e$ and $\|e\| \leq \eta\|r\|$, $r^T Js \leq -(1-\eta)\|r\|^2$; the existing line search verifies actual decrease. An existing BiCGStab-backend probe of this frozen system diverges to true relative residual 1.73×10^{103} after 1200 iterations (2402 JVPs, 401.8 seconds), with no accepted Newton update. Optional augmented GMRES retains three previous-cycle correction directions; CPU and CUDA manufactured controls pass, and a small nonlinear solve agrees between backends to 2.57×10^{-13} in sampled fields. Its additional workspace raises the 240-grid allocation bound to 61.27 GiB, within 64 GiB. On the frozen failing system it reaches relative residual 0.471 after 2400 iterations (421.4 seconds), still above 0.1, with no Newton update. Thus neither alternative resolves the large-system stagnation. After the projection correction, the failed GMRES residual instead has 98.53% momentum energy spread over modes 0–7; modes 0–1 carry only 26.57%. A separate evaluation of the retained 240-grid nonlinear state finds interior residual concentration: the outermost eight radial and polar layers carry only 4.99×10^{-6} and 3.29×10^{-8} of total squared residual, respectively. These observations do not support an endpoint-only repair or another Krylov-method sweep. The same-checkpoint CPU/CUDA residual comparison gives $\|F_{\mathrm{CPU}} - F_{\mathrm{CUDA}}\|_2 = 2.11 \times 10^{-13}$, only 1.42×10^{-4} of the CUDA residual norm 1.48×10^{-9}. Backend-specific evaluation differences therefore do not account for the dominant stalled residual; common discretization or precision errors remain possible. Neither backend is assumed to be exact. Reanalysis of these saved arrays finds 93.60% of

weighted residual energy in the last eight radial DCT modes and 78.56% in the last eight polar modes; median degrees are 238 and 477 on 240 and 480 nodes. Both backends agree. This identifies grid-scale stagnation and motivates improving the spectral operator's low-order preconditioner, without establishing the cause or justifying residual filtering. The opt-in spectral radial preconditioner now uses exact polynomial radial differentiation in the existing dense blocks, retaining polar finite differences and endpoint conditions. The residual and Jacobian-vector product are unchanged; the default backend and Bowen–York path are unaffected. Compact CPU assembly avoids dense entries in per-point sparse containers. Added host derivative/map tables require $16n_r^2 + 24n_r$ bytes, included in the aggregate memory guard. Manufactured radial controls give worst scaled errors 7.00×10^{-13} on CPU and 3.66×10^{-14} on CUDA. The small binary requires five Newton steps and 48 Krylov iterations (previously 53), with CPU/CUDA sampled-field difference 2.39×10^{-13}. These controls qualify the implementation, not fine-grid effectiveness.

The final frozen-state GPU probe (Perlmutter job 59426128) uses the same 240-grid checkpoint, GMRES restart 64, relative target 0.1, and budget 2400. It fails with true relative residual 0.78667352, 2439 Jacobian-vector products, and 448.51 seconds of diagnostic solve time. No accepted Newton update or improved physical solution follows. The option remains disabled by default; it is not recommended as a demonstrated remedy for the fine-grid stall. Evidence is retained in `validation/trumpet/c4-experiment/axis-tau/polar-exact/spectral-radial/gpu-v1/probe.json`.

Final nonlinear and physical criteria remain unchanged. Binary acceptance remains open.

Retained alternative	Decisive outcome
Factored C4 unknowns	Manufactured controls pass; fine binary fails line search. Tighter forcing, longer backtracking and coarse continuation do not cure it.
Preconditioner scaling	Two prototypes fail the small coupled control; rejected.
Stable scalar-source assembly	Reduces measured angular source roundoff, but fine replay still needs damping 1/2048 with negligible residual decrease.
Monotone FD preconditioner	Small solve improves; fine replay still needs 1/1024 damping with negligible residual decrease.
Local polar endpoint rows	Fine linear solve fails after 2400 iterations; exact radial rows alone also fail (relative residual 0.0162).
Exact radial and polar rows	Fine first linear solve passes in 6 iterations; full nonlinear solve converges; coarse physical error still requires refinement.

These experiments are not a new backend benchmark campaign. Full configurations, failed states and precision/transfer diagnostics remain under `validation / trumpet / c4-experiment/`; none are discarded or promoted to physical acceptance. The original C2 default and successful historical data remain available.

The retained boost map is $(\lambda, \kappa) = (0.03, 3.5)$. Radial refinement from 160 to 256 improved its errors by factors 2–6. Stronger focusing, $\lambda = 0.003$, worsened both independent errors and radial tails and is rejected. Radial-512 mapped-operator controls pass at normalized error 5.49×10^{-10}. After the original timeout, the reduced-factor retry completed: relative to radial 256, near-hole Hamiltonian/momentum RMS errors improve by factors 8.09/3.95 and bulk errors by 4.18/3.39. Extrapolated energy changes by 5.48×10^{-6} relatively. These improvements do not meet the physical bounds; the diagnostic horizon check also fails.

V. HORIZON MEASUREMENTS AND COMPUTATIONAL COST

The earlier pre-tau trumpet moderate 256-grid checkpoint passes import and horizon angular/quadrature checks, but remains diagnostic because of its constraint sequence. Its component irreducible masses are 0.5887780114 and 0.3935773863.

The spin-target diagnostic checkpoint passes orders $12, 16, 20$ and an independent quadrature increase. Each component has $M_H = 0.500017851900$, $M_{\mathrm{irr}} = 0.377763585476$, and coordinate $\chi = 0.989929520988$, with expansion RMS 5.98×10^{-9}. Both retained surfaces enclose the modified regions. These are coordinate rotation-integral spins, not approximate-Killing-vector estimates. Reusing passed harmonic shapes strictly as initial guesses reduces order-12/16 runs from 671/1227 seconds to 35.5/50.4 seconds; all geometry and acceptance checks are recomputed.

For boosted horizons an optional spatial chart $x = c + J(y - c)$, $J = I - vv^T/(1+\sqrt{1-v^2})$, makes the exact isolated seed surface spherical. Tensors and derivatives are pulled back consistently; physical rotation generators become $J^{-1}E_iJ$. This changes no slice. The isolated $\Gamma = 10$ control passes orders $4, 8, 12$: relative area error $\leq 7.22 \times 10^{-11}$, shape error $\leq 1.13 \times 10^{-10}$, and expansion RMS $\leq 4.78 \times 10^{-11}$. The solved binary instead stalls near 1.8×10^{-5} at both orders 8 and 16; it has no qualified horizon result. At radial 512 the order-8 expansion floor falls to 9.49×10^{-6} after 1000 iterations, still above 10^{-7}. Import succeeds with ADM/Z4c relative error 4.39×10^{-16}. The driver stops at the failed surface; its area does not supply a qualified irreducible mass for separation calibration.

Necessary production solves, not a matched-accuracy benchmark, took 94.2 seconds (315 seconds total, 4.87 GiB peak process RSS) for the listed spin case and 161.3 seconds (273 seconds total, 3.53 GiB RSS) for the listed boost case. Host RSS excludes GPU allocation. The radial-512 run retains the existing 64 GiB aggregate guard; its conservative estimate is 63.74 GiB. The completed radial-512 retry took 3119.6 seconds in the nonlinear solve (3294.6 seconds total, 5.59 GiB peak host RSS), with 10 Newton steps and 352 Krylov iterations. It used the original column-wise transfer construction.

The longer radial-512 attempt exposed a host factor-construction bottleneck: a live stack snapshot found active threads in the modal preconditioner's column-wise triangular solves. The qualified opt-in implementation constructs $A^{-1}\,\mathrm{diag}(U)$ with two matrix-right-hand-side triangular solves, retaining the same LU factors, pivots and block recurrence. On a 512-row pivoted control, the original-matrix residual is 7.41×10^{-16} and the old/new transfer difference is 5.11×10^{-16}. Necessary kernel timings on Perlmutter are 0.556 versus 0.107 seconds (a factor of 5.19); **this is not a whole-solver speedup measurement**. A local small-solve field/gradient comparison agrees within 1.31×10^{-16}, and the separate CUDA axisymmetric control passes. The option remains off by default. A continuation

command pins the tested image and preserves the radial-512 configuration. It was not launched because the original run completed; future necessary solves can use the qualified optimization. Evidence is in `validation/trumpet/batched-transfer/` and `validation/trumpet/axial-factors/stack-snapshot.txt`.

VI. REPRODUCIBILITY AND REMAINING WORK

The native and consumer branches are `codex / hispid-trumpet` and `codex / hispid-trumpet-pgen`; neither is merged. Detailed evidence, failed attempts, hashes and executable identities are under `validation / trumpet/`, with the audit trail in `docs/TRUMPET_IMPLEMENTATION.md`. Key records are `convergence/assessment.json`, `spin-focused-map / horizon160-warm/`, `affine-horizon / exact / horizons-v4/`, and `axisymmetric/radial-comparison.json`. Commands are retained beside each run. The original performance report is preserved without extending its campaign.

Completion still requires an accepted moderate physical-convergence sequence, resolution of the extreme binary constraint errors, qualified boost-binary horizon measurements and measured-$M_{\rm irr}$ separation calibration, followed by a final evidence audit. Pending runs and failed gates are not counted as validation. This report is an interim snapshot, not completion of the goal.

References: Dennison–Baumgarte–Montero (2014); Ruchlin et al., high-spin/high-boost puncture data; boosted trumpet construction reference.